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File 1: Malware Case Study - "SHADOWBYTE" Ransomware Intrusion

CASE STUDY: SHADOWBYTE RANSOMWARE INCIDENT

Incident ID: IR-2026-089

Date of Attack: 2026-04-10 to 2026-04-18

Target: Pacifica General Hospital (PGH) - 450-bed facility

Sector: Healthcare

ATTACK TIMELINE (as reconstructed from logs and EDR):

2026-04-10 08:45: An HR administrator at PGH receives an email from "resume-submissions@jobs-portal.net" with subject "Application: Senior IT Analyst
Position - John Vasquez". Attachment: "John_Vasquez_Resume.zip"

2026-04-10 08:52: Administrator extracts the zip file. Inside is "John_Vasquez_Resume.js" (JavaScript file). Windows SmartScreen shows no warning.

2026-04-10 08:53: User double-clicks the .js file. A PowerShell window flashes briefly and disappears. No visible change occurs.

2026-04-10 08:54: The JavaScript executes PowerShell to download from <http://5.188.86.45/update.ps1>. Script writes file to C:\Users\Public\svhost.exe (note: misspelled "svchost").

2026-04-10 08:55: Malware modifies registry key: HKLM\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection\DisableRealtimeMonitoring to value 1 (disables Defender).

2026-04-10 08:56: Malware creates scheduled task named "AdobeUpdateTask" to run svhost.exe every 6 hours.

2026-04-10 09:00: Malware runs ``net localgroup administrators /add temp_admin`` and ``net user temp_admin P@ssw0rd123! /add`` creating a new local admin account.

2026-04-10 09:15: Malware uses BloodHound (legitimate AD exploration tool) to map domain structure: ``SharpHound.exe -c All --domain pacifica.local``

2026-04-10 to 2026-04-12: No logs - malware dormant during weekend.

2026-04-13 01:30: Malware uses impacket (Python-based tool) to move laterally to file server FS02 at 10.10.20.15 using stolen domain admin credentials.

2026-04-13 01:45: On FS02, malware runs ``vssadmin delete shadows /all /quiet`` - deletes all volume shadow copies (prevents recovery).

2026-04-13 02:00: Malware begins encrypting files with extensions: .dcm (medical images), .pdf, .xlsx, .mdb (databases). Uses AES-256 encryption. Appends extension ".shadowbyte" to encrypted files.

2026-04-13 02:30: Malware drops ransom note "README_TO_DECRYPT.txt" in every folder:
"We have encrypted your patient records and medical imaging data. Pay 75 Bitcoin (value ~\$2.5M USD) to wallet address 1A2b3C4d5E6f7G8h9I0j to receive decryption key.
You have 72 hours. Do not contact law enforcement."

2026-04-13 03:00: Malware exfiltrates 15 GB of unencrypted patient data (as extortion leverage) to 5.188.86.45 via HTTPS POST before encryption completes.

2026-04-13 05:00: Hospital staff arrive to find files inaccessible. Ransom note displayed on FS02 and 40+ workstations.

2026-04-13 06:30: Hospital initiates incident response. FBI Cyber Task Force notified.

2026-04-14 to 2026-04-18: Recovery from backups (36 hours). ~8 GB of data exfiltrated before containment confirmed. No ransom paid.

TECHNICAL INDICATORS:

- Domain: jobs-portal.net (registered 2026-04-05 via NameCheap)
- IP: 5.188.86.45 (hosted in Russia, known ransomware infrastructure)
- File hashes: John_Vasquez_Resume.js (SHA256: 4a5b6c7d8e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b)
- PowerShell download URL: http://5.188.86.45/update.ps1
- Encrypted file extension: .shadowbyte
- Bitcoin wallet: 1A2b3C4d5E6f7G8h9I0j
- Ransomware family: ShadowByte (new variant of Ryuk/Ragnar Locker family)

File 2: MITRE ATT&CK Matrix Template - SHADOWBYTE

MITRE ATT&CK MATRIX - SHADOWBYTE RANSOMWARE

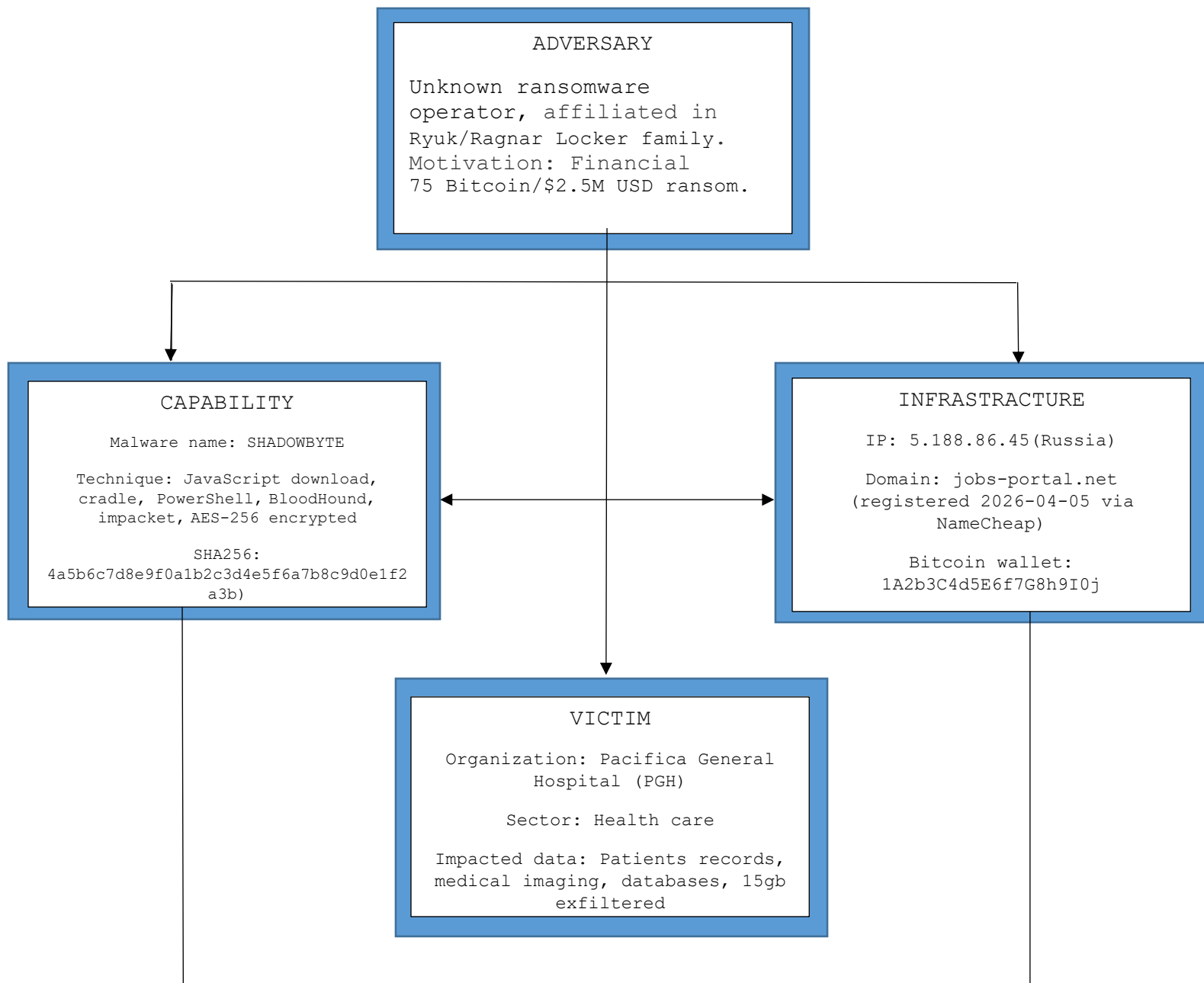
Fill in specific Technique ID and Description for each Tactic observed

Tactic	Technique ID	Technique Name	Specific Evidence from Case Study
Reconnaissance			
Resource Development	T1583.001	Acquire Infrastructure: Domains	jobs-portal.net (registered 2026-04-05 via NameCheap)
Initial Access	T1566.001	Phishing: Spear phishing Attachment	receives an email from "resume-submissions@jobs-portal.net" with subject

			"Application: Senior IT Analyst Position - John Vasquez". Attachment: "John Vasquez Resume.zip"
Execution	T1059.005	Command and Scripting Interpreter: Visual Basic/JScript	JavaScript executes PowerShell to download from http://5.188.86.45/update.ps1 .
Persistence	T1053.005	Scheduled Task/Job: Scheduled Task	Malware creates scheduled task named "AdobeUpdateTask" to run svchost.exe every 6 hours.
Privilege Escalation	T1136.001	Create Account: Local Account	runs `net localgroup administrators /add temp_admin` and `net user temp_admin P@ssw0rd123! /add` creating a new local admin account.
Defense Evasion	T1490	Inhibit System Recovery	runs `vssadmin delete shadows /all /quiet` - deletes all volume shadow copies.
Credential Access	T1552.001	Unsecured Credentials: Credentials in Files	uses impacket (Python-based tool) to move laterally to file server FS02 at 10.10.20.15 using stolen domain admin credentials.
Discovery	T1482	Domain Trust Discovery	`SharpHound.exe -c All --domain pacifica.local`
Lateral Movement	T1021.002	Remote Services: SMB/Windows Admin Shares	uses impacket (Python-based tool) to move laterally to file server FS02 at 10.10.20.15
Collection	T1005	Data from Local System	begins encrypting files with extensions: .dcm (medical images), .pdf, .xlsx, .mdb (databases).
Command and Control	T1071.001	Application Layer Protocol: Web Protocols	exfiltrates 15 GB of unencrypted patient data (as extortion leverage) to 5.188.86.45 via HTTPS POST
Exfiltration	T1041	Exfiltration Over C2 Channel	exfiltrates 15 GB of unencrypted patient data to 5.188.86.45 via HTTPS POST before encryption completes.
Impact	T1485	Data Destruction	runs `vssadmin delete shadows /all /quiet` - deletes all volume shadow copies (prevents recovery).

NOTE: Not all tactics may be present. Leave blank if no evidence exists for that tactic.
Minimum 12 techniques required across all tactics.

DIAMOND MODEL - SILENTBANKER INCIDENT



META-FEATURES:

Timestamp (When): [2026-04-10 to 2026-04-18]

Phase (Stage of attack): Impact/Encryption (exfiltration phase)

Result (What happened): 40+ workstations and 8GB data exfiltrated, no ransom paid, recovery from backups

Direction (Data flow): Victim (PGH) to Adversary (5.188.86.45) for exfiltration; Adversary to Victim for encryption.

KILL CHAIN MAPPING (add to Diamond):

Reconnaissance → Weaponization → Delivery → Exploitation → Installation → C2 → Actions

Kill Chain Phase	Evidence from Case Study
Reconnaissance	Not explicitly observed (assumed)
Weaponization	John Vasquez Resume.js
Delivery	Email with attachment John_Vasquez_Resume.zip on April 10, 08:145
Exploitation	User double-clicking .j file on April 10, 08:53
Installation	writes file to C:\Users\Public\svhost.exe on April 10, 08:54
Command & Control	exfiltrates 15 GB of unencrypted patient data (as extortion leverage) to 5.188.86.45 via HTTPS POST on April 13, 03:00
Actions on Objectives	Domain mapping, lateral movement to FS02, shadow copy deletion, AES-256 encryption, ransom note drop, 15 GB data exfiltration April 10, 09:15 - April 13, 03:00

FRAMEWORK CROSS-MAPPING TABLE

Diamond Element	MITRE Tactic	Kill Chain Phase	Specific Evidence
Capability: JavaScript execution	Execution (T1059.005)	Exploitation	User double-clicks the .js file
Capability: PowerShell download	Execution(T1059.001)	Installation	JavaScript executes PowerShell to download from http://5.188.86.45/update.ps1.
Infrastructure: 5.188.86.45	Command and Control (T1071.001)	Command & Control	exfiltrates 15 GB of patient data to 5.188.86.45 via HTTPS POST
Victim: PGH	Impact (T1486)	Actions on Objectives	AES-256 encryption. Appends extension '.shadowbyte
Capability: vssadmin	Defense Evasion (T1490)	Actions on Objectives	vssadmin delete shadows /all /quiet - deletes all volume shadow copies

Final Review - Verify:

✓MITRE matrix has minimum 12 filled tactics

✓Each filled tactic has Technique ID, Name, AND exact evidence quote

- ✓Diamond diagram has all 4 corners filled
- ✓Diamond meta-features (4) are all filled
- ✓Kill Chain mapping has all 7 phases (some may say "not observed")
- ✓Cross-mapping table has at least 4 rows